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JEE Mains 2020 Jan Chapter wise Question Bank

SHM

Q1

Length of a simple pendulum is 25.0 cm and time of 40 oscillation is 50 sec. If resolution of stop watch is 1 sec then accuracy is g is (in %)

एक सरल लोलक की लम्बाई 25.0 cm है तथा 40 दोलों के लिए 50 sec लगते हैं। यदि घड़ी के लिए अल्पतमांक 1 sec है तो g में प्रतिशत त्रुटि ज्ञात करें।

(1) 2.4

(2) 3.4

(3) 4.4

(4) 5.4

8th Jan Evening

Sol

(3)

$$\frac{\Delta T}{T} = \frac{1}{2} \left(\frac{\Delta g}{g} + \frac{\Delta L}{L} \right)$$

$$\frac{\Delta g}{g} = \frac{2\Delta T}{T} + \frac{\Delta L}{L} ; \quad = 2 \left(\frac{1}{50} \right) + \frac{0.1}{25.0}$$

$$= 4.4 \%$$

Q2

A loop of radius R and mass m is placed in a uniform magnetic field B with its plane perpendicular to the field. Current I is flowing in it. Now loop is slightly rotated about its diameter and released. Find time period of oscillation.

R त्रिज्या तथा m द्रव्यमान का लूप नियत चुम्बकीय क्षेत्र B के लम्बवत् रखा है। इसमें धारा I बह रही है। अब लूप को उसके व्यास के अनुदिश अल्प कोण से घुमाते हैं तो छोड़ने के पश्चात् आवर्तकाल ज्ञात करो—

$$(1) 2\sqrt{\frac{\pi m}{IB}}$$

$$(2) \sqrt{\frac{2\pi m}{IB}}$$

$$(3) 2\sqrt{\frac{m}{\pi IB}}$$

$$(4) \sqrt{\frac{m}{\pi IB}}$$

9th Jan Evening

Sol

SHM**JEE Mains 2020 Jan Chapter wise Question Bank****(2)**

$$\tau = MB \sin \theta = I \alpha$$

$$\pi R^2 I B \theta = \frac{m R^2}{2} \alpha$$

$$\omega = \sqrt{\frac{2 \pi I B}{m}} = \frac{2 \pi}{T}$$

$$T = \sqrt{\frac{2 \pi m}{I B}}$$



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